Sample Exam

Medical Image Analysis

Student name:____

Immatriculation number:____

Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions the slices are acquired one after the other in a pre-defined direction and stacked into a 3D volume; in axial acquisitions the slices lie in the transverse plane. Voxel sizes are given as in-plane (two values) and through-plane (one value, the slice thickness), and the field-of-view is image size $\times$ voxel size. A *dynamic* (4D) acquisition repeats the whole volume a number of times, one volume per time point.

   An axial dynamic acquisition is performed with in-plane voxel size 1.5×1.5 mm^2 and 128 pixels in both directions, through-plane voxel size 5.0 mm with 30 slices, and 200 time points acquired at a temporal resolution of 2.5 s per volume.

   (a) What is the field-of-view in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

   (b) How many voxels does the complete 4D data set contain, and how long does the acquisition take?

   (c) A second scan of the same patient is acquired at $1.0 \times 1.0 \times 1.0$ mm^3 isotropic. To compare measurements between the two scans, which property of the images has to be matched? Give the resize factors needed to bring one volume of the dynamic acquisition onto that grid, and the resulting image size.

2. Otsu's method selects a threshold for foreground–background separation automatically from the image histogram. Let p_1, p_2 be the numbers of pixels assigned to the two groups by a candidate threshold and σ_1^2, σ_2^2 the intensity variances within those groups. Which of the two criteria below is the one Otsu's method uses? Please circle the correct choice.

$$\min_{\tau} \; p_1\sigma_1^2 + p_2\sigma_2^2 \qquad\qquad \max_{\tau} \; p_1\sigma_1^2 + p_2\sigma_2^2$$

3. Denoising can be written either as an energy minimisation or as a maximum-a-posteriori estimate. J is the observed noisy image and I the image we want to recover.

$$\max_{I} \; \log P(J \mid I) + \log P(I) \qquad\qquad \min_{I} \; \lambda\, D(I,J) + R(I)$$

   (a) Please indicate which term corresponds to which term in the two formulations.

   (b) For total variation denoising the data term is $D(I,J) = \|I - J\|_2^2$ and the corresponding likelihood is Gaussian, $P(J \mid I) = \mathcal{N}(J;\, I,\, \sigma^2)$. What is λ in terms of σ, and what does that mean in words?

4. Please indicate whether the statement is true or false by circling the correct choice in each of the following. A denotes a binary image viewed as a set of pixels, B a structural element, and B_x the structural element translated to x.

   (a) True False The dilation of A by B is the set of all x for which B_x has a non-empty intersection with A.

(b) True False The opening of A by B is a dilation followed by an erosion, and it fills small gaps and holes.

(c) True False Morphological operations are linear and shift-invariant.

(d) True False Applying a median filter to a label map cannot introduce labels that were not already present in it.

5. In landmark-based registration a set of corresponding landmark pairs is used to determine the transformation parameters. How many corresponding landmark pairs are needed for the system to be exactly determined when the transformation model is (i) a 2D affine transformation, and (ii) a 3D affine transformation? Explain the counting.

6. Optimisation in image registration is difficult, and the objective function generally has local minima that a gradient-based optimiser can get stuck in. Please describe the two practical strategies presented in the lecture for dealing with this, and briefly explain why each of them helps. You do not need to write down the equations.

7. Convolutional networks reduce the spatial dimensions of their feature maps either with pooling or with strided convolutions. Which of the following statements is **inaccurate**?

   ○ (A) Max pooling provides partial local translation invariance: the position of the maximum can move within the pooling window without changing the pooled value.

   ○ (B) Pooling is applied to each channel separately and has no learnable parameters at all.

   ○ (C) A strided convolution reduces the spatial dimensions and, like pooling, buys a degree of translation invariance.

   ○ (D) Average pooling is a linear operation, whereas max pooling is not.

8. Three standard measures are used to judge how well a statistical shape model has embedded the training data into a lower-dimensional space. Which of the following statements is **incorrect**?
   - ○ (A) Compactness measures how much of the total variance is explained by the first t modes, $C(t) = \sum_{i \le t} \lambda_i / \lambda_{total}$; a compact model needs few parameters to generate new shape instances.
   - ○ (B) Generalisation measures the ability of the model to represent instances that were not in the training set, and is estimated with leave-one-out reconstruction.
   - ○ (C) Specificity measures how accurately the model reconstructs the training shapes themselves; a perfectly specific model reproduces every training instance exactly.
   - ○ (D) Principal component analysis assumes that the data follow a Gaussian distribution, and produces an orthogonal lower-dimensional coordinate system that maximises the variance along each successive axis.

9. Which of the following statements about positional encoding and the structure of a transformer layer is **incorrect**?
   - ○ (A) Without positional encoding a transformer is equivariant to permutations of its input tokens, so "I bought an apple watch" and "watch an apple I bought" cannot be distinguished.
   - ○ (B) An ideal positional encoding should give a unique representation for each position, be bounded, generalise to sequences longer than those seen in training, and encode the relative positions of tokens.
   - ○ (C) In the sinusoidal scheme the positional vector is concatenated to the token embedding, which doubles the dimension of the representation entering the first transformer layer.
   - ○ (D) A transformer layer consists of multi-head self-attention followed by a per-token MLP, each of the two wrapped in a residual connection and followed by layer normalisation.

10. The magnitude image of an MR acquisition is formed from the real and imaginary channels as $J(x) = \sqrt{(I_r(x) + \epsilon)^2 + (I_i(x) + \eta)^2}$, where $\epsilon, \eta \sim \mathcal{N}(0, \sigma)$ are the noise in the two channels. What kind of noise does the resulting magnitude image have, and why does it matter?
    - ○ (A) Additive Gaussian noise, since ϵ and η are both Gaussian and the operations applied to them are continuous.
    - ○ (B) Multiplicative speckle noise, of the same type as in ultrasound and optical coherence tomography.
    - ○ (C) Rician noise — taking the magnitude makes the noise non-Gaussian and signal-dependent, so denoising methods derived from an additive Gaussian model are mis-specified, particularly in low-signal regions.
    - ○ (D) Poisson noise, of the same type as the quantum noise in low-dose CT.
    - ○ (E) No noise at all, because the two independent noise terms cancel in the magnitude operation.

11. Consider a classification network that takes grey-scale input images of size 60×60 with one input channel. All convolutions are *valid* (no padding, $P = 0$) with stride $S = 1$, and each is followed by max pooling with $K = 2 \times 2$ and $S = 2$:
    - **Conv1**: 12 kernels with $K = 5 \times 5$ **Pool1**: 2×2, $S = 2$

- **Conv2**: 24 kernels with $K = 5 \times 5$ **Pool2**: 2×2, $S = 2$
- **Conv3**: 48 kernels with $K = 3 \times 3$ **Pool3**: 2×2, $S = 2$

Recall that a valid convolution gives $N_l = N_{l-1} - k_1 + 1$. Circle True or False:

(a) True False The output of Pool3 has a spatial size of 5×5.

(b) True False Each neuron in Pool3 has a global receptive field covering the entire 60×60 input image.

(c) True False The total number of convolutional kernel weights (excluding the biases) is $5 \cdot 5 \cdot 1 \cdot 12 + 5 \cdot 5 \cdot 12 \cdot 24 + 3 \cdot 3 \cdot 24 \cdot 48 = 17\,868$.

(d) True False Replacing the three max-pooling layers by 2×2 convolutions with stride 2 would leave the total number of learnable parameters unchanged, because pooling and strided convolution both simply downsample.

12. Which of the following statements on domain shift, adaptation and uncertainty are correct (multiple answers are possible)?

   ◯ (A) Acquisition shift is a change in $P_D(X \mid Z)$ — scanner, resolution, contrast, protocol — and is the kind of dataset shift encountered most often in medical imaging.

   ◯ (B) Prevalence shift is a change in $P_D(Y \mid X)$, caused for instance by differences in annotation policy or in the experience of the annotator.

   ◯ (C) The Expected Calibration Error measures the average gap between the confidence a model assigns to its predictions and the accuracy it actually achieves at that confidence.

   ◯ (D) Adapting only the batch-normalisation parameters of a network is not possible, because batch normalisation has no learnable parameters.

   ◯ (E) A deep ensemble approximates $p(\theta \mid D, M)$ by training the same architecture several times from different random initialisations, on the assumption that the different runs land in different modes of the parameter posterior.